GATE

Computer Science & Information Technology

Computer Networks - 1

Routing

DPP: 1

Q1 Link state routing is based on _________

(A) Dijkstra's algorithm

(B) Bellman-Ford algorithm

(C) Floyd-Warshall algorithm

(D) Flooding

Q2 Distance vector routing is based on __________ .

(A) Dijkstra's algorithm

(B) Bellman-Ford algorithm

(C) Floyd-Warshall algorithm

(D) Flooding

Q3 __________ is a link state routing protocol .

(A) RIP (Routing Information Protocol)

(B) OSPF (Open Shortest Path First)

(C) BGP (Border Gateway Protocol)

(D) IS-IS (Intermediate System to Intermediate System

Q4 __________ is a distance vector routing protocol .

(A) RIP (Routing Information Protocol)

(B) OSPF (Open Shortest Path First)

(C) BGP (Border Gateway Protocol)

(D) IS-IS (Intermediate System to Intermediate System

Q5 'Count to infinity' problem is associated with which routing algorithms ?

(A) Distance Vector Routing

(B) Link State Routing

(C) Path Vector Routing

(D) None of the above

Q6

Identify correct statement regarding 'Link state' routing algorithm .

(A) Dijkstra Algorithm executed by any one router

(B) Dijkstra Algorithm executed by a particular routers

(C) Dijkstra Algorithm executed by all the routers

(D) No need to run Dijkstra Algorithm explicitly

Q7 Consider a network of three routers A, B, and C, where B is connected to A and C with distance 4 and 5 respectively.

Identify correct distance vector of router B after one iteration (round of update) of the distance vector routing algorithm?

(A) A = 5, B = 0, C = 4

(B) A = 4, B = 0, C = ∞

(C) A = ∞, B = 0, C = 5

(D) A = 4, B = 0, C = 5

Q8 Consider a network of five routers : P, Q, R, S and T, uses 'Distance Vector' routing algorithm. Every router maintain distance to other router in the above mentioned routers sequence. Router Q is directly connected with router P and S with distance 4 and 2 respectively. If distance vector of P : [0, 4, 3, 5, 2] and distance vector of S : [6, 2, 4, 0, 3] are arrived at router Q, then what should be router Q updated distance vector?

(A) [4, 0, 6, 2, 6]

(B) [4, 0, 7, 2, 5]

(C) [4, 0, 6, 2, 5]

(D) [4, 0, 7, 2, 6]

Q9

Consider a network of five routers : A, B, C, D and E, uses 'Distance Vector' routing algorithm. Every router maintain distance to other router in the above mentioned routers sequence. Router C is directly connected with router A and B with distance 3 and 5 respectively. If distance vector of A : [0, 5, 3, 4, 2] and distance vector of B : [6, 0, 4, 5, 3] are arrived at router C, then what should be router C updated distance vector

(A) [0, 3, 5, ∞, ∞]
(B) [3, 5, 0, ∞, ∞]
(C) [3, 5, 0, 10, 6]
(D) [3, 5, 0, 7, 5]

Q10 Which of the following statements is/are correct regarding distance vector and link state routing protocols ?

(A) Distance vector routing protocol is prone to routing loops than link state routing protocol.
(B) Link state routing protocols generally consume more bandwidth than distance vector routing protocols.
(C) Distance vector routing protocols converge faster than link state routing protocols.
(D) Link state routing protocols ensure that each router possesses a consistent and identical view of the network topology

Answer Key

Q1	(A)	Q6	(C)
Q2	(B)	Q7	(D)
Q3	(B)	Q8	(C)
Q4	(A)	Q9	(D)
Q5	(A)	Q10	(A, B, D)

Hints & Solutions

Q1 Text Solution:

Link state routing is based on Dijkstra's algorithm

Q2 Text Solution:

Distance vector routing is based on Bellman-Ford algorithm

Q3 Text Solution:

OSPF (Open Shortest Path First)

Q4 Text Solution:

RIP (Routing Information Protocol)

Q5 Text Solution:

Distance Vector Routing

Q6 Text Solution:

Dijkstra Algorithm executed by all the routers

Q7 Text Solution:

A = 4, B = 0, C = 5

Q8 Text Solution:

P Q R S T

Q —4→ P : [0, 4, 3, 5, 2]

Q —2→ P : [0, 4, 3, 5, 2]

P Q R S T

Q : [4, 0, 6, 2,5]

P Q S S S

[4, 0, 6, 2, 5]

Q9 Text Solution:

C —3→ A : [0, 5, 3, 4, 2]

C —5→ B : [6, 0, 4, 5, 3]

A B C D E

C : [3, 5, 0, 7, 5]

A B C A A

[3, 5, 0, 7, 5]

Q10 Text Solution:

Distance vector routing protocol is prone to routing loops than link state routing protocol. **(True)**

Link state routing protocols generally consume more bandwidth than distance vector routing protocols. **(True)**

Distance vector routing protocols converge faster than link state routing protocols. **(False)**

Link state routing protocols ensure that each router possesses a consistent and identical view of the network topology. **(True)**